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Examples: *Taenia solium* (tape worm), *Fasciola hepatica* (liver fluke), *Schistosoma* (blood fluke).

2.3.5 Phylum: Aschelminthes (Round Worms)

(G. *Askes* –cavity; *helminths* – worms)

Previously called Nematoda, this phylum is now named as Aschelminthes. The body of these worms is circular (round) in cross section and hence are called round worms. They are free living or parasitic on aquatic and terrestrial plants and animals. They are bilaterally symmetrical, triploblastic and pseudocoelomate animals with organ system level of organisation. The body is unsegmented and covered by a transparent, tough and protective collagenous layer called cuticle. The alimentary canal is complete with a well developed mouth, muscular pharynx and anus. Excretory system consists of rennet glands. Sexes are separate; and exhibit sexual dimorphism; often females are longer than males. Fertilisation is internal; majority are oviparous (e.g. *Ascaris*) few are ovoviviparous (*Wuchereria*). Development may be direct or indirect.

Examples. *Ascaris lumbricoides* (round worm), *Enterobius vermicularis* (Pin worm), *Wuchereria bancrofti* (filarial worm), *Ancylostomaa deuodenale* (hook worm) (Figure 2.13).

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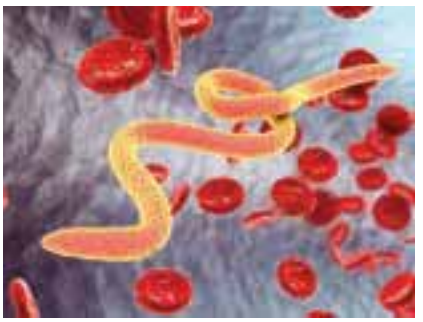
2.3.6 Phylum: Annelida (Segmented worm)

(L. *annulus* -a ring, and G. *edios*- form)

Annelids were the first segmented animals to evolve. They are aquatic or terrestrial, free living but some are parasitic. They are triploblastic, bilaterally symmetrical, schizocoelomates and exhibit organ system level of body organisation. The coelom with coelomic fluid creates a hydrostatic skeleton and aids in locomotion. Their elongated body is metamerically segmented and the body surface is divided into segment or metameres. Internally the segments are divided from one another by partitions called septa. This phenomenon is known as metamerism. The longitudinal and circular muscles in the body wall help in locomotion. Aquatic annelids like *Nereis* have lateral appendages called parapodia, which help in swimming. Chitinous setae



Ascaris



Filarial worm



Hook worm

Figure 2.13 Examples of Aschelminthes